

DURGA MOORTHY

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EDUCATION

BACHELOR OF BIOMEDICAL ENGINEERING

Vanderbilt University, Dean's List

May 2027

GPA: 3.75 • Relevant Coursework: Rapid Prototyping, Biomechanics, Systems Physiology, Quantitative Methods: Statistical Analysis

SKILLS

DESIGN TOOLS: Autodesk Fusion • Autodesk Inventor • Adobe Photoshop • Slicing Software • Figma • Webflow • GitHub Pages

PROGRAMMING TOOLS: Python • Java • HTML • CSS • Google Apps Script • MATLAB

OTHER: Fluent English and Tamil • Basic Spanish • Machine Shop Skills

WORK EXPERIENCE

VANDERBILT INSTITUTE FOR SURGERY AND ENGINEERING (VISE)

May 2025 - Present

Summer Fellowship

- Collected and processed intraoperative AR-guided gaze data; collaborated with urology surgeons in the operating room and during phantom studies. Continuously refined Unity-based HoloLens 2 system, optimized protocols, and resolved technical issues to ensure accurate real-time tracking.
- Analyzed gaze data using Python (scikit-learn) for statistical and machine learning methods. Presented findings at the **VISE symposium and VUSE poster session**.
- Prepared funding reports; currently drafting papers for peer-reviewed journals and 2026 conference submissions.
- Mentored a high school student in introductory video processing, data analysis and statistics.

MACHINE AUTOMATION PERCEPTION AND LEARNING (MAPLE) LAB

Jan 2025 - Present

Student Research

- Developed a data organization and processing system using OpenCV, LosslessCut and Google Apps Script to sort and process thousands of surgical videos for ARPA-H (Advanced Research Projects Agency for Health)

VETERANS AFFAIRS HOSPITAL

May 2023 - Aug 2023

Medical Pathology Program

- Selected as one of four students for an immersive hospital internship, gaining hands-on experience in dissection, autopsy procedures, surgical protocols, medical morgue protocols, and medical terminology.
- Participated in a rotations-based program, shadowing OR cases and diagnostic conferences while working alongside medical residents in podiatry, biomedical implants, radiology, endoscopy, and pathology.

THE CODER SCHOOL

May 2022 - June 2024

Computer Science Teacher

- Designed adaptive coding curricula for Python, Java, Scratch, and game-based platforms for a diverse group of students (ages 7–85) in computer science, adjusting methods to individual learning styles.
- Created AP Computer Science (APCSA/APCSP) prep materials, combining real-world programming challenges with exam-focused content.

LEADERSHIP & ACTIVITIES

Societies:

- **Society of Biomedical Engineers & Women of VISE:** Actively participated in networking events, talks, and site visits, and recruited new members through tabling and outreach activities.
- **Society of Women Engineers – Executive Board:** Coordinated professional and social events including launching a big-little mentorship program, organizing coffee chats, and facilitating bonding activities.

Leadership:

- **Upperclassmen Resident Advisor (Current):** Organize community events, attend weekly staff meetings, trained in conflict mediation and emergency response.
- **TEDx Speaker & Program Director:** Delivered a TEDx talk on childhood alopecia; directed a program with 1M+ viewers and led speaker workshops and student teams.
- **FIRST Robotics – Senior Board Member:** Oversaw design strategy and part fabrication; developed compliant intake and spring-based elevator systems, optimized rotary joint for 3D printing, and trained underclassmen to improve team efficiency.
- **Varsity Water Polo – Team Captain:** 2x Junior Olympian; earned Outstanding Academic All-American Scholar and St. Francis Scholar Athlete Award. **Certified Lifeguard (American Red Cross CPR/AED)**